NeuroChip Functional Electrophysiology Report

Project: Drug screening 2026   Sample: Organoid001   Recording: fs363-org0

Event QC	Stability	Phenotype axis	Ref. deviation
100/100 PASS	PASS	P33.3	P2.2

Recording and quality control

Duration	400.0 s
Channels / detected events	8 / 4,217
Active channel fraction	87.5%
Inactive channel candidates	1
Review / excluded candidates	0 / 0
Firing stability CV	0.105
Last-to-first activity change	+0.058 log2
Raw-voltage QC	Not available from event input

Activity, criticality, and connectivity

Mean firing rate	1.3178 Hz
Network burst rate	39.000 /min
Mean STTC (20 ms)	0.1188
Avalanche rate	140.850 /min
Avalanche mean size	4.491 events
Avalanche branching ratio	1.002
Bias-corrected mutual information	0.005243 bits
Bias-corrected transfer entropy	0.002347 bits
Pairwise predictive GC score	0.026908

Largest reference-deviation Shapley contributions

isi_median_s	+0.00276 (scaled value +0.15)
dominant_channel_fraction	+0.00134 (scaled value -0.14)
active_channel_fraction	+0.00107 (scaled value -0.33)
network_clustering	+0.00105 (scaled value -0.05)
network_density	+0.00083 (scaled value -0.15)
isi_cv_mean	+0.00018 (scaled value -0.13)

Interpretation boundary

This report evaluates detected extracellular spike events. A silent event channel is an inactive candidate, not a confirmed dead electrode. Raw-voltage noise, line interference, saturation, baseline drift, and waveform morphology require raw voltage. Transfer entropy and Granger scores describe statistical information flow or predictive direction and do not prove synaptic causality. Phenotype-axis and reference-deviation percentiles are relative to the 45-recording forebrain-organoid cohort. The phenotype axis is not a health or maturity score, and reference deviation is not a disease, efficacy, toxicity, or safety probability.